

Cybersecurity Internship Report

Week 1 Submission

Reconnaissance and Network Traffic Analysis using Nmap & Wireshark

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Week: 1

Submission Date: 28 June 2026

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1. Objective

The objective of this task was to perform basic reconnaissance on a vulnerable virtual machine (Metasploitable2), identify open ports and running services using Nmap, monitor network traffic using Wireshark, and understand common security risks associated with exposed services.

2. Environment Setup

Host Operating System

- Windows 11

Virtualization Software

- Oracle VirtualBox

Target Virtual Machine

- Metasploitable2

Network Configuration

- Adapter 1: NAT
- Adapter 2: Host-only Adapter

Target Hostname

- metasploitable

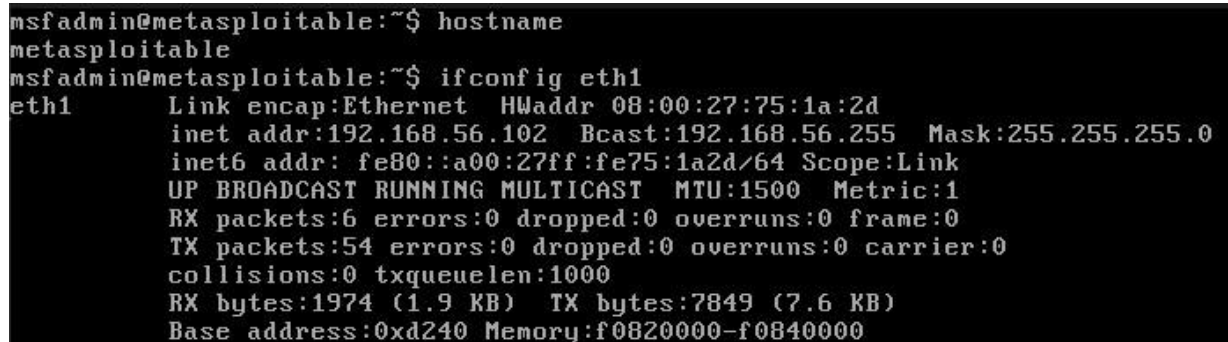
Target IP Address

- 192.168.56.102
-

Screenshot 1

VM terminal showing

- Hostname
- Ifconfig eth1



```
msfadmin@metasploitable:~$ hostname
metasploitable
msfadmin@metasploitable:~$ ifconfig eth1
eth1      Link encap:Ethernet  HWaddr 08:00:27:75:1a:2d
          inet addr:192.168.56.102  Bcast:192.168.56.255  Mask:255.255.255.0
          inet6 addr: fe80::a00:27ff:fe75:1a2d/64  Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:6 errors:0 dropped:0 overruns:0 frame:0
          TX packets:54 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:1974 (1.9 KB)  TX bytes:7849 (7.6 KB)
          Base address:0xd240  Memory:f0820000-f0840000
```

Figure 1: Metasploitable2 hostname and IP configuration

3. Tools Used

Tool	Purpose
VirtualBox	Virtualization Platform
Metasploitable2	Vulnerable Linux Machine
Nmap	Network Reconnaissance
Wireshark	Packet Analysis
Windows PowerShell	Executing Scan Commands

4. Methodology

The following steps were performed:

1. Started Metasploitable2 VM.
 2. Verified IP address.
 3. Verified connectivity using Ping.
 4. Performed basic Nmap scan.
 5. Performed Service Version Detection.
 6. Performed OS Detection.
 7. Performed Aggressive Scan.
 8. Captured network traffic using Wireshark.
 9. Filtered HTTP, FTP and SYN packets.
 10. Documented findings.
-

5. Reconnaissance using Nmap

Step 1 - Connectivity Test

The target machine was first verified using the Ping command.

Command:

```
ping 192.168.56.102
```

The host responded successfully with 0% packet loss.

Screenshot 2

```
PS C:\Users\gg> ping 192.168.56.102

Pinging 192.168.56.102 with 32 bytes of data:
Reply from 192.168.56.102: bytes=32 time=11ms TTL=64
Reply from 192.168.56.102: bytes=32 time<1ms TTL=64
Reply from 192.168.56.102: bytes=32 time<1ms TTL=64
Reply from 192.168.56.102: bytes=32 time<1ms TTL=64

Ping statistics for 192.168.56.102:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 11ms, Average = 2ms
```

Figure 2: Successful connectivity with target machine

Step 2 - Basic Port Scan

Command

```
nmap 192.168.56.102
```

Purpose

- Discover live host
- Identify open ports

Results

Multiple TCP ports were found open including FTP, SSH, HTTP, SMB, MySQL and PostgreSQL.

Screenshot 3

```
PS C:\Users\gg> nmap 192.168.56.102
Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-27 11:14 +0530
Nmap scan report for 192.168.56.102
Host is up (0.0058s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8009/tcp  open  ajp13
8180/tcp  open  unknown
MAC Address: 08:00:27:75:1A:2D (Oracle VirtualBox virtual NIC)
```

Figure 3: Basic Port Scan

Step 3 - Service Version Detection

Command

```
nmap -sV 192.168.56.102
```

Purpose

Identify versions of running services.

Important findings

- vsFTPD 2.3.4
 - Apache 2.2.8
 - OpenSSH 4.7
 - Samba 3.x
 - MySQL 5.0.51a
 - PostgreSQL 8.3
-

Screenshot 4

```
Nmap done: 1 IP address (1 host up) scanned in 1.48 seconds
PS C:\Users\gg> nmap -sV 192.168.56.102
Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-27 11:16 +0530
Nmap scan report for 192.168.56.102
Host is up (0.0066s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE        VERSION
21/tcp    open  ftp            vsftpd 2.3.4
22/tcp    open  ssh            OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet         Linux telnetd
25/tcp    open  tcpwrapped
53/tcp    open  domain         ISC BIND 9.4.2
80/tcp    open  http           Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind        2 (RPC #100000)
139/tcp   open  netbios-ssn    Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn    Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec           netkit-rsh rshcd
513/tcp   open  login?
514/tcp   open  tcpwrapped
1099/tcp  open  java-rmi       GNU Classpath grmiregistry
1524/tcp  open  bindshell      Metasploitable root shell
2049/tcp  open  nfs            2-4 (RPC #100003)
2121/tcp  open  ftp            ProFTPD 1.3.1
3306/tcp  open  mysql          MySQL 5.0.51a-3ubuntu5
5432/tcp  open  postgresql     PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp  open  vnc            VNC (protocol 3.3)
6000/tcp  open  X11            (access denied)
6667/tcp  open  irc            UnrealIRCd
8009/tcp  open  ajp13          Apache Jserv (Protocol v1.3)
8180/tcp  open  http           Apache Tomcat/Coyote JSP engine 1.1
MAC Address: 08:00:27:75:1A:2D (Oracle VirtualBox virtual NIC)
Service Info: Host: irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 13.35 seconds
```

Figure 4: Service Version Detection

Step 4 - Operating System Detection

Command

```
nmap -O 192.168.56.102
```

Purpose

Identify target operating system.

Result

Nmap detected Linux Kernel 2.6.x.

Screenshot 5

```
C:\Windows\System32>nmap -O 192.168.56.102
Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-27 11:17 +0530
Nmap scan report for 192.168.56.102
Host is up (0.0013s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8009/tcp  open  ajp13
8180/tcp  open  unknown
MAC Address: 08:00:27:75:1A:2D (Oracle VirtualBox virtual NIC)
Device type: general purpose
Running: Linux 2.6.X
OS CPE: cpe:/o:linux:linux_kernel:2.6
OS details: Linux 2.6.9 - 2.6.33
Network Distance: 1 hop

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 2.61 seconds
```

Figure 5: Operating System Detection

Step 5 - Aggressive Scan

Command

```
nmap -A 192.168.56.102
```

Purpose

Collect detailed service information, scripts, versions, operating system details and traceroute.

Major findings

- FTP Anonymous Login Enabled
- Apache HTTP Server
- Samba Shares
- MySQL
- PostgreSQL
- VNC
- IRC
- RPC Services
- HTTP Headers
- SMB Information

Screenshot 6

```
C:\Windows\System32>nmap -A 192.168.56.102
Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-27 11:21 +0530
Nmap scan report for 192.168.56.102
Host is up (0.0018s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
|_ftp-anon: Anonymous FTP login allowed (FTP code 230)
|_ftp-syst:
|_STAT:
|_FTP server status:
|_  Connected to 192.168.56.1
|_  Logged in as ftp
|_  TYPE: ASCII
|_  No session bandwidth limit
|_  Session timeout in seconds is 300
|_  Control connection is plain text
|_  Data connections will be plain text
|_  vsFTPd 2.3.4 - secure, fast, stable
|_End of status
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
|_ssh-hostkey:
|_  1024 60:0f:cf:e1:c0:5f:6a:74:d6:90:24:fa:c4:d5:6c:cd (DSA)
|_  2048 56:56:24:0f:21:1d:de:a7:2b:ae:61:b1:24:3d:e8:f3 (RSA)
23/tcp    open  telnet       Linux telnet
25/tcp    open  tcpwrapped
|_smtp-command: Couldn't establish connection on port 25
53/tcp    open  domain       ISC BIND 9.4.2
|_dns-nsid:
|_  bind.version: 9.4.2
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)
|_http-title: Metasploitable2 - Linux
|_http-server-header: Apache/2.2.8 (Ubuntu) DAV/2
111/tcp   open  rpcbind      2 (RPC #100000)
|_rpcinfo:
|_  program version  port/proto  service
|_  100000  2           111/tcp    rpcbind
|_  100000  2           111/udp    rpcbind
|_  100003  2,3,4       2049/tcp   nfs
|_  100003  2,3,4       2049/udp   nfs
|_  100005  1,2,3       43320/udp  mountd
|_  100005  1,2,3       49829/tcp  mountd
```

Figure 6: Aggressive Scan Results (Part 1)

Screenshot 7

```
100005 1,2,3 49829/tcp mountd
100021 1,3,4 44419/udp nlockmgr
100021 1,3,4 49762/tcp nlockmgr
100024 1 34659/udp status
100024 1 36220/tcp status
139/tcp open netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp open netbios-ssn Samba smbd 3.0.20-Debian (workgroup: WORKGROUP)
512/tcp open exec netkit-rsh rshd
513/tcp open login?
514/tcp open tcpwrapped
1099/tcp open java-rmi GNU Classpath grmiregistry
1524/tcp open bindshell Metasploitable root shell
2049/tcp open nfs 2-4 (RPC #100003)
2121/tcp open ftp ProFTPD 1.3.1
3306/tcp open mysql MySQL 5.0.51a-3ubuntu5
mysql-info:
  Protocol: 10
  Version: 5.0.51a-3ubuntu5
  Thread ID: 9
  Capabilities flags: 43564
  Some Capabilities: SupportsCompression, SupportsTransactions, LongColumnFlag, Speaks41ProtocolNew, ConnectWithDatabase, SwitchToSSLAfterHandshake, Support41Auth
  Status: Autocommit
  Salt: mB?Vte-hmLUD6}8<dbB
5432/tcp open postgresql PostgreSQL DB 8.3.0 - 8.3.7
ssl-cert: Subject: commonName=ubuntu804-base.localdomain/organizationName=OCOSA/stateOrProvinceName=There is no such thing outside US/countryName=XX
  Not valid before: 2010-03-17T14:07:45
  Not valid after: 2010-04-16T14:07:45
  _ssl-date: 2026-06-27T05:52:29+00:00; +7s from scanner time.
5900/tcp open vnc VNC (protocol 3.3)
vnc-info:
  Protocol version: 3.3
  Security types:
  VNC Authentication (2)
6000/tcp open X11 (access denied)
6667/tcp open irc UnrealIRCd
irc-info:
  users: 1
  servers: 1
  lusers: 1
  lservers: 0
  server: irc.Metasploitable.LAN
  version: Unreal3.2.8.1. irc.Metasploitable.LAN
  uptime: 0 days, 0:24:53
```

Figure 7: Aggressive Scan Results (Part 2)

Screenshot 8

```
uptime: 0 days, 0:24:53
source ident: nmap
source host: AF59FDED.97684684.FFFA6D49.IP
error: Closing Link: jtunizqdm[192.168.56.1] (Quit: jtunizqdm)
8009/tcp open ajp13 Apache Jserv (Protocol v1.3)
ajp-methods: Failed to get a valid response for the OPTION request
8180/tcp open http Apache Tomcat/Coyote JSP engine 1.1
http-favicon: Apache Tomcat
http-server-header: Apache-Coyote/1.1
http-title: Apache Tomcat/5.5
MAC Address: 08:00:27:75:1A:2D (Oracle VirtualBox virtual NIC)
Device type: general purpose
Running: Linux 2.6.X
OS CPE: cpe:/o:linux:linux_kernel:2.6
OS details: Linux 2.6.9 - 2.6.33
Network Distance: 1 hop
Service Info: Host: irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Host script results:
_smb2-time: Protocol negotiation failed (SMB2)
smb-os-discovery:
  OS: Unix (Samba 3.0.20-Debian)
  Computer name: metasploitable
  NetBIOS computer name:
  Domain name: localdomain
  FQDN: metasploitable.localdomain
  System time: 2026-06-27T01:52:06-04:00
smb-security-mode:
  account_used: <blank>
  authentication_level: user
  challenge_response: supported
  message_signing: disabled (dangerous, but default)
_nbstat: NetBIOS name: METASPLOITABLE, NetBIOS user: <unknown>, NetBIOS MAC: <unknown> (unknown)
_clock-skew: mean: 1h20m06s, deviation: 2h18m34s, median: 6s

TRACEROUTE
HOP RTT ADDRESS
1 1.81 ms 192.168.56.102

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 69.51 seconds
```

Figure 8: Aggressive Scan Results (Part 3)

6. Network Traffic Analysis using Wireshark

Traffic was captured while performing Ping and Nmap scans.

Different protocol filters were applied to inspect network communication.

HTTP Analysis

Filter Used

http

Observation

HTTP GET and POST requests were captured successfully.

The communication occurred in plaintext over port 80.

Screenshot 9

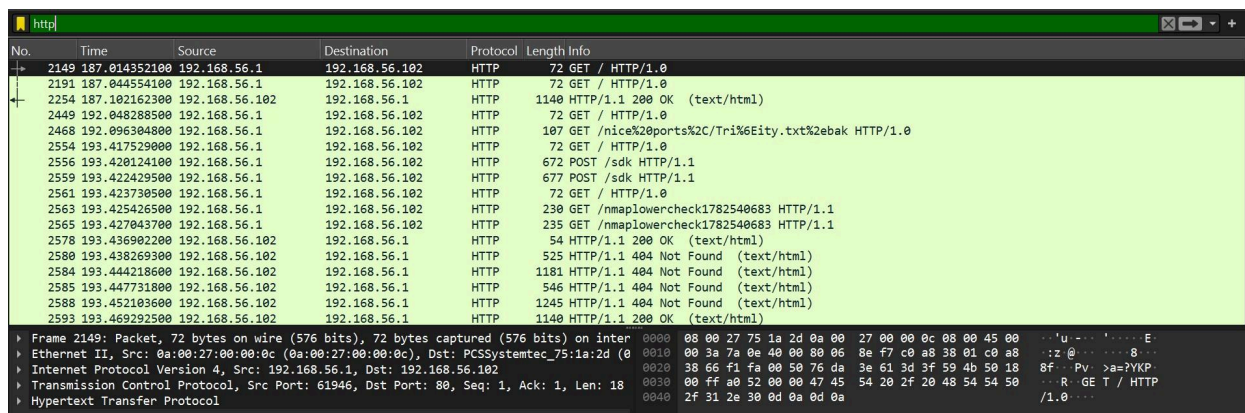


Figure 9: HTTP Traffic Analysis

FTP Analysis

Filter Used

ftp

Observation

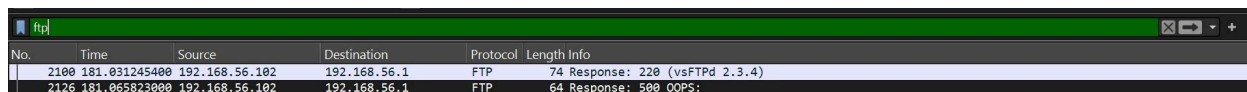
FTP responses were captured.

The server reported:

vsFTPD 2.3.4

indicating an outdated FTP service.

Screenshot 10



A screenshot of a Wireshark packet capture window. The title bar shows 'ftp'. The packet list pane displays two packets:

No.	Time	Source	Destination	Protocol	Length	Info
2100	181.031245480	192.168.56.102	192.168.56.1	FTP	74	Response: 220 (vsFTPD 2.3.4)
2126	181.065823000	192.168.56.102	192.168.56.1	FTP	64	Response: 500 OOPS:

Figure 10: FTP Traffic Analysis

TCP SYN Analysis

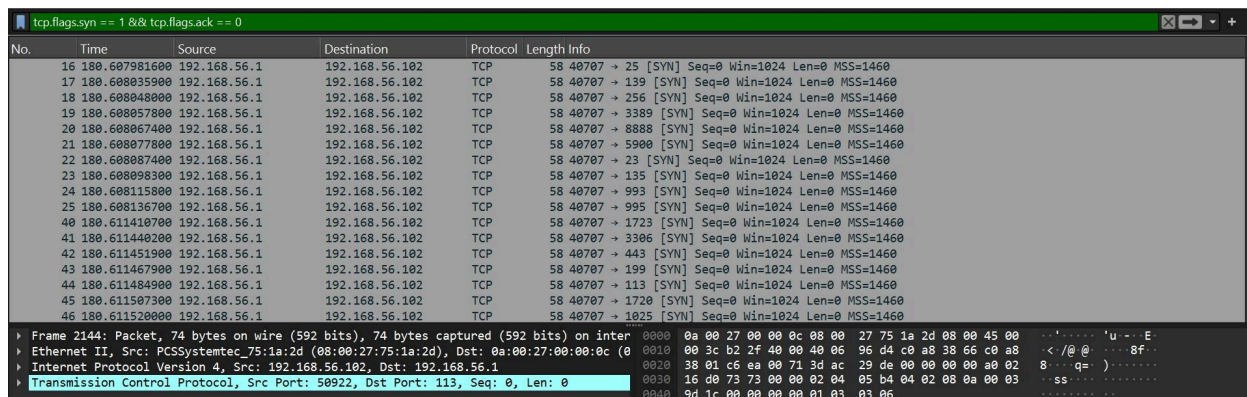
Filter Used

`tcp.flags.syn == 1 && tcp.flags.ack == 0`

Observation

A large number of SYN packets were observed during Nmap scanning, representing TCP connection attempts to multiple ports.

Screenshot 11



A screenshot of a Wireshark packet capture window. The title bar shows 'tcp.flags.syn == 1 && tcp.flags.ack == 0'. The packet list pane displays a list of SYN packets:

No.	Time	Source	Destination	Protocol	Length	Info
16	180.607981600	192.168.56.1	192.168.56.102	TCP	58	40707 → 25 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
17	180.608035900	192.168.56.1	192.168.56.102	TCP	58	40707 → 139 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
18	180.608048000	192.168.56.1	192.168.56.102	TCP	58	40707 → 256 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
19	180.608057800	192.168.56.1	192.168.56.102	TCP	58	40707 → 3389 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
20	180.608067400	192.168.56.1	192.168.56.102	TCP	58	40707 → 8888 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
21	180.608077800	192.168.56.1	192.168.56.102	TCP	58	40707 → 5900 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
22	180.608087400	192.168.56.1	192.168.56.102	TCP	58	40707 → 23 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
23	180.608098300	192.168.56.1	192.168.56.102	TCP	58	40707 → 135 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
24	180.608115800	192.168.56.1	192.168.56.102	TCP	58	40707 → 993 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
25	180.608136700	192.168.56.1	192.168.56.102	TCP	58	40707 → 995 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
40	180.611410700	192.168.56.1	192.168.56.102	TCP	58	40707 → 1723 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
41	180.611440200	192.168.56.1	192.168.56.102	TCP	58	40707 → 3306 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
42	180.611451900	192.168.56.1	192.168.56.102	TCP	58	40707 → 443 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
43	180.611467900	192.168.56.1	192.168.56.102	TCP	58	40707 → 199 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
44	180.611484900	192.168.56.1	192.168.56.102	TCP	58	40707 → 113 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
45	180.611507300	192.168.56.1	192.168.56.102	TCP	58	40707 → 1720 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
46	180.611520000	192.168.56.1	192.168.56.102	TCP	58	40707 → 1025 [SYN] Seq=0 Win=1024 Len=0 MSS=1460

The packet details pane shows the following information for the selected packet (No. 46):

- Frame 2144: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface 0
- Ethernet II, Src: PCSSystemtec_75:1a:2d (08:00:27:75:1a:2d), Dst: 0a:00:27:00:00:0c (08:00:27:00:00:0c)
- Internet Protocol Version 4, Src: 192.168.56.102, Dst: 192.168.56.1
- Transmission Control Protocol, Src Port: 50922, Dst Port: 113, Seq: 0, Len: 0

Figure 11: TCP SYN Packet Analysis

7. Observations

Open Ports

- 21 FTP
- 22 SSH
- 23 Telnet
- 25 SMTP
- 53 DNS
- 80 HTTP
- 111 RPC
- 139 NetBIOS
- 445 SMB
- 3306 MySQL
- 5432 PostgreSQL
- 5900 VNC
- 8180 Apache Tomcat
- Several additional RPC and network services

Important Running Services

- vsFTPD 2.3.4
- Apache HTTP Server 2.2.8
- OpenSSH 4.7
- Samba 3.x
- MySQL 5.0.51a
- PostgreSQL 8.3

Potential Risks

- Outdated FTP Server
- Anonymous FTP Login
- Open Telnet Service
- Multiple Exposed Network Services
- Plain HTTP Communication
- Weak SMB Configuration

8. Threat Summary & Impact

Threat

vsFTPD 2.3.4

Risk

Remote Code Execution

Threat	Risk
Anonymous FTP	Unauthorized Access
Telnet	Plaintext Credentials
HTTP	Unencrypted Communication
Samba	Information Disclosure
Multiple Open Ports	Increased Attack Surface

9. Recommendations

- Disable Telnet.
 - Upgrade outdated FTP server.
 - Disable Anonymous FTP login.
 - Replace HTTP with HTTPS.
 - Close unnecessary ports.
 - Apply operating system security updates.
 - Enable firewall rules.
 - Use strong authentication mechanisms.
-

10. Conclusion

This exercise provided hands-on experience with basic penetration testing reconnaissance and network monitoring. Using Nmap, the target machine was successfully enumerated, revealing multiple outdated and vulnerable services. Wireshark traffic analysis demonstrated how HTTP, FTP, and TCP SYN packets appear during reconnaissance activities. The practical exercise improved understanding of network scanning techniques, packet analysis, and the importance of identifying exposed services before exploitation.